Camera-Only Multi-View BEV Occupancy Estimation Under Diverse Weather Conditions

Pranav Jain, Avantika Rattan, Venkatakrishnan Amruthur Kesavan
University of Michigan, Ann Arbor, US

UNIVERSITY OF MICHIGAN

Motivation

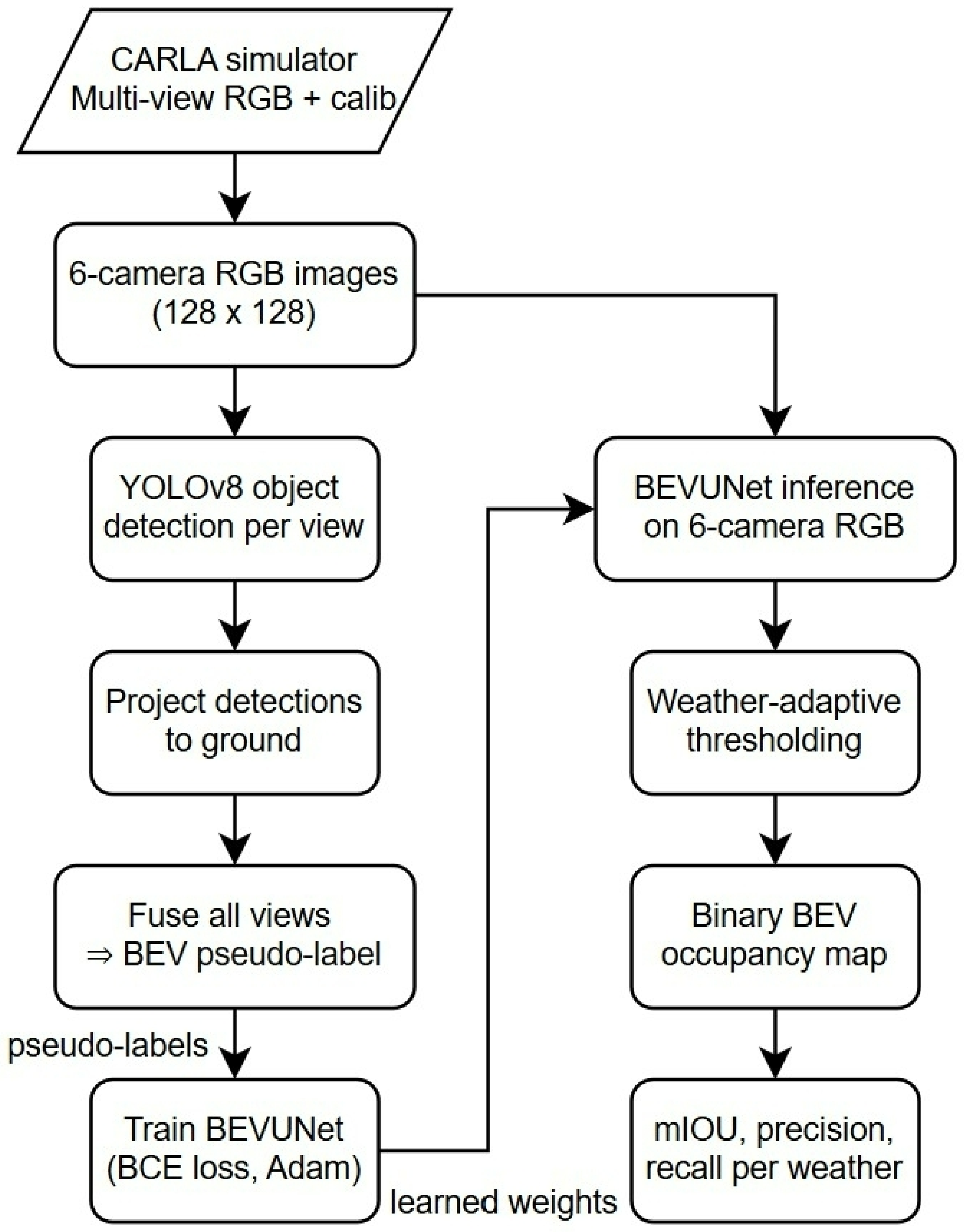
- BEV requires multimodal sensor inputs –
  - LiDAR, radar & cameras.

- Costly & extensive annotations on dataset. Limits scalability.
- Camera-only BEV is cheaper.

- Areas of Improvements:
  - Explicity Depth Information
  - Robust Multi-view Feature Aggregation

Dataset Generation

- Steps:
  - Used Carla simulator to simulate real scenarios on the road across 3 weather conditions.
  - Placed 6 cameras in the following positions: front, back, front-left, front-right, left and right
  - Used YOLOv8 to generate BEV pseudo labels
  - Transformed image data positions onto the 3D BEV Space
  - Finally obtained RGB images, pseudo labels and objects oriented on the BEV space.
    Foggy, Rainy and clear images displayed.



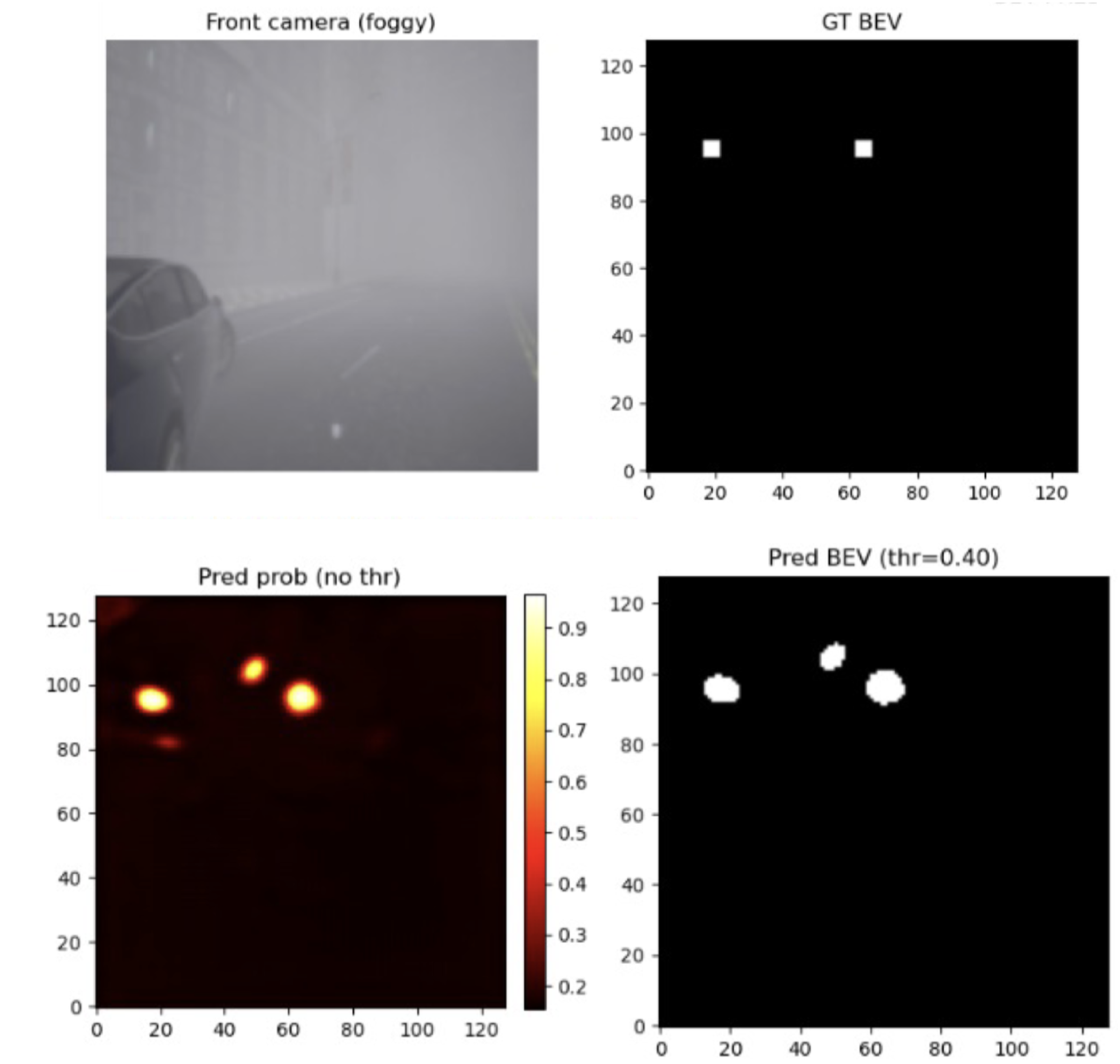
Methodology

- Steps:
  - Employed a U-Net–style encoder–decoder architecture
  - Fixed BEV grid - a 60m x 60m region centered around ego
  - Training is done based on Binary Cross-Entropy with Logits
  - Inference is performed by converting to probabilities using a sigmoid activation function and binarizing the results using weather-specific thresholds τw.
  - Higher thresholds are applied in clear scenes to suppress false positives, while lower thresholds are used in rainy and foggy conditions to preserve recall, thereby improving robustness under adverse weather conditions.

Result

Weather	mIoU	Precision	Recall	mAP
Foggy	0.193	0.205	0.868	0.178
Rainy	0.180	0.183	0.950	0.174
Clear	0.366	0.393	0.869	0.342

Method	Weather	mAP	mIoU
PETR [5]	Clear	0.370	-
PETRv2 [6]	Clear	0.421	-
BEVFormer [3]	Clear	0.416	0.487
BEVDet [4]	Clear	0.397	-
BEVDet4D [4]	Clear	0.396	-
DA-BEV [10]	Clear	0.203	0.356
DA-BEV [10]	Rainy	0.110	-
Ours (BEVUNet)	Clear	0.342	0.366
Ours (BEVUNet)	Rainy	0.174	0.180
Ours (BEVUNet)	Foggy	0.178	0.193



Future Work

- Pseudo - label quality can be enhanced using ensemble object detectors
- Incorporating depth aware features
- Including temporal awareness
- Extending to multi class BEV prediction

[1] H. Li, Y. Zhao, J. Zhong, B. Wang, C. Sun, and F. Sun, "Delving into the Secrets of BEV 3D Object Detection in Autonomous Driving: A Comprehensive Survey," Authorea Preprints, 2024. [2] J. Philion and S. Fidler, "Lift, Splat, Shoot: Encoding Images from Arbitrary Camera Rigs by Implicitly Unprojecting to 3D," in Proceedings of the European Conference on Computer Vision (ECCV), 2020